Let's now consider the improper integral $\int_1^\infty \frac{1}{x^p}\,dx$.

We have already seen that this improper integral converges if $p = 2$ and diverges if $p = 1$. What about for other values of $p$?

**Theorem**

*The improper integral* $\int_1^\infty \frac{1}{x^p}\,dx = \frac{1}{p-1}$ *when* $p > 1$. *However it is divergent when* $p \leq 1$.

We have to consider only the case $p \neq 1$. By the definition,

$$\int_1^\infty \frac{1}{x^p}\,dx = \lim_{t\to\infty} \int_1^t \frac{1}{x^p}\,dx = \lim_{t\to\infty} \left[\frac{x^{-p+1}}{-p+1}\right]_1^t$$

$$= \frac{1}{1-p} \cdot \lim_{t\to\infty} \left(t^{1-p} - 1\right) = \begin{cases} \infty, & \text{if } p < 1, \\ \dfrac{1}{p-1}, & \text{if } p > 1. \end{cases}$$

## (2) Improper integrals of type II

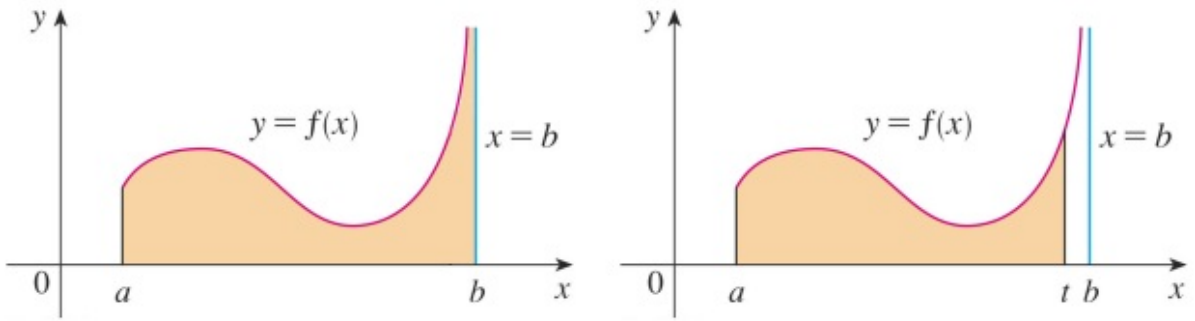
Attention! *You should be able to recognize the improper integrals of this type (Type II) and know how to evaluate them.*

Under this type the function has a vertical asymptote at one of the two endpoints, or at the inside of the interval.

As an example, consider a function $f(x)$ which is continuous on the half open interval $[a, b)$, but which has a vertical asymptote at $x = b$. This region is an unbounded region because it stretches vertically without bound.

So how can we find the area of such a region?

Notice that for every $t < b$, the function is continuous on the closed interval $[a, t]$ and so the definite integral $\int_a^t f(x)\,dx$ will exist. So we can define the area of this unbounded region to be the limit of the bounded region $\int_a^t f(x)\,dx$ as $t$ approaches the vertical asymptote $b$.




(The left region is unbounded, while the right region is bounded).

There are three types of an improper integral of type II:

(a) If $f$ is continuous on the interval $[a, b)$ and has a vertical asymptote at $x = b$, then

$$\int_a^b f(x)\,dx = \lim_{t \to b^-} \int_a^t f(x)\,dx,$$

provided that the one-side limit on the right exists and is finite.

(b) If $f$ is continuous on the interval $(a, b]$ and has a vertical asymptote at $x = a$, then

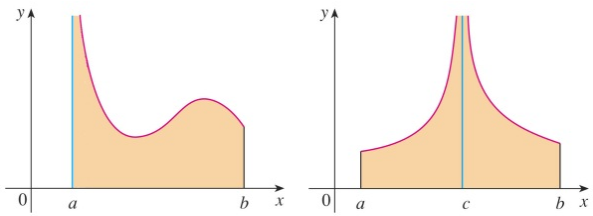
$$\int_a^b f(x)\,dx = \lim_{t \to a^+} \int_t^b f(x)\,dx,$$

provided that the one-side limit on the right exists and is finite.

(c) If $f$ is continuous on the intervals $[a, c)$ and $(c, b]$, and $f(x)$ has a vertical asymptote at $x = c$ then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$

provided that both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ exist and are finite.




(Improper integral of type II (b) on the left and of type II (c) on the right).

**Some remarks**.

1) Again we call an improper integral of type II convergent if the associated limit(s) exist and are finite. Otherwise we call the improper integral divergent.

2) In case (c) above we define

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

Since $x = c$ is a vertical asymptote, both $\int_a^c f(x)\,dx$, $\int_c^b f(x)\,dx$ are also improper integrals (and of type II). So we may write

$$\begin{aligned}\int_a^b f(x)\,dx &= \int_a^c f(x)\,dx + \int_c^b f(x)\,dx \\ &= \left(\lim_{t\to c^-}\int_a^t f(x)\,dx\right) + \left(\lim_{t\to c^+}\int_t^b f(x)\,dx\right).\end{aligned}$$

If one or both of $\lim_{t\to c^-} \int_a^t f(x)\,dx$ and $\lim_{t\to c^+} \int_t^b f(x)\,dx$ are divergent, then the improper integral $\int_a^b f(x)\,dx$ is divergent.

3) If there are more than one vertical asymptote in the range $[a, b]$, say $x = c_1, c_2, \cdots, c_k$, we can calculate the improper integral $\int_a^b f(x)\,dx$ as follows:

$$\int_a^b f(x)\,dx = \int_a^{c_1} f(x)\,dx + \int_{c_1}^{c_2} f(x)\,dx + \cdots + \int_{c_k}^b f(x)\,dx.$$

We require that all of $\int_a^{c_1} f(x)\,dx, \int_{c_1}^{c_2} f(x)\,dx, \ldots, \int_{c_k}^b f(x)\,dx$ be convergent.

4) Sometimes in the question we are not told that $\int_a^b f(x)\,dx$ is an improper integral.

It is important to first check quickly if there are any vertical asymptotes in the interval $[a, b]$.

If there are, we must treat it as an improper integral of type II.

Evaluating $\int_a^b f(x)\,dx$ directly may result in a different answer.

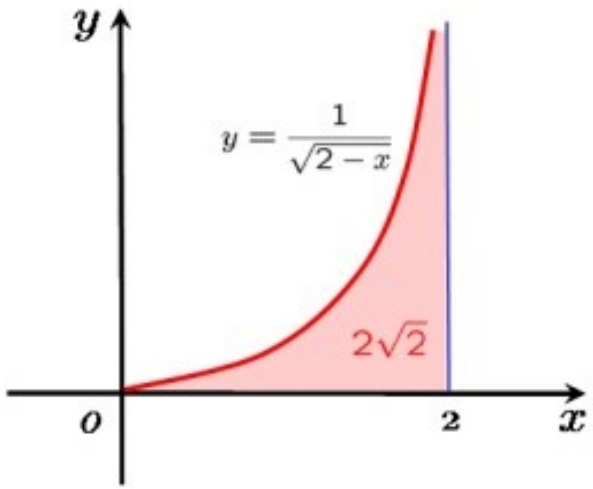
## Example

Evaluate $\displaystyle\int_0^2 \frac{1}{\sqrt{2-x}}\,dx$.

Solution. Quickly checking reveals that $x = 2$ is a vertical asymptote and the function is continuous elsewhere $[0, 2)$. Hence we must treat this as an improper integral of type II.

$$\begin{aligned}\int_0^2 \frac{1}{\sqrt{2-x}}\,dx &= \lim_{t\to 2^-}\int_0^t \frac{1}{\sqrt{2-x}}\,dx = \lim_{t\to 2^-}\int_0^t (2-x)^{-\frac{1}{2}}\,dx \\ &= \lim_{t\to 2^-}\left[\frac{(2-x)^{\frac{1}{2}}}{\frac{1}{2}\cdot(-1)}\right]_0^t = \lim_{t\to 2^-}\left[-2\sqrt{2-x}\right]_0^t \\ &= -2\lim_{t\to 2^-}\sqrt{2-t} + 2\sqrt{2} = 0 + 2\sqrt{2} = 2\sqrt{2}.\end{aligned}$$

Thus we have the unbounded region of type II with finite area $2\sqrt{2}$.



## (3) Comparison theorem

Attention! *You should be able to apply the comparison theorem to improper integrals of type I.*

- Sometimes it is impossible to find the exact value of an improper integral. However we still would like to know whether it is convergent or divergent.

- In this case we can compare the given improper integral with another one (that we know how to evaluate).

## Theorem (Comparison theorem for improper integrals of type I)

*Suppose that $f(x)$ and $g(x)$ are continuous functions on $[a, \infty)$ with $f(x) \geq g(x) \geq 0$ for all $x \in [a, \infty)$. Then,*

(a) *If $\int_a^\infty f(x)\,dx$ is convergent, then $\int_a^\infty g(x)\,dx$ is convergent.*

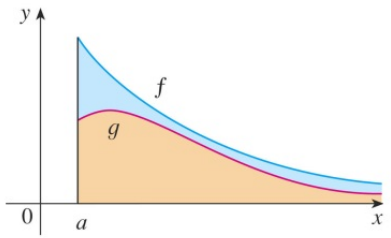
(b) *If $\int_a^\infty g(x)\,dx$ is divergent, then $\int_a^\infty f(x)\,dx$ is divergent.*

## Some comments.

(a) It is important to check that both $f(x)$ and $g(x)$ are non-negative. It does not work if $f(x)$ or $g(x)$ are negative.

(b) The converse of the theorem may not be true. For instance if the bigger integral $\int_a^\infty f(x)\,dx$ is divergent then the smaller integral $\int_a^\infty g(x)\,dx$ might or might not diverge. Similarly, if the smaller integral $\int_a^\infty g(x)\,dx$ is convergent then the larger integral $\int_a^\infty f(x)\,dx$ might or might not converge.

(c) The theorem can be seen intuitively from the following figure:




($f(x) \geq g(x) \geq 0$ for all $x \geq a$).

(d) The comparison theorem can help us to prove that a given improper integral $\int_a^\infty g(x)\,dx$ is convergent, by comparing with a larger improper integral $\int_a^\infty f(x)\,dx = A$ which also converges.

However it does not tell us the exact value of the limit $\int_a^\infty g(x)\,dx$ (even though we can conclude that it is some finite number $B$).

The only thing we can say for sure is that $B \leq A$, but the comparison theorem does not tell us that $A = B$ or any other value.

To find the value of $B = \int_a^\infty g(x)\,dx$ we must evaluate some other way, or use numerical integration to approximate $B$.

## Example

Determine if $\displaystyle\int_1^\infty \frac{1+e^{-x}}{x}\,dx$ is convergent.

Solution. We already know that $\displaystyle\int_1^\infty \frac{1}{x}\,dx$ is divergent.

So we should find some inequality involving $\dfrac{1+e^{-x}}{x}$ and $\dfrac{1}{x}$.

Let's bound the numerator. Clearly, $1+e^{-x} > 1$, for all $x \geq 1$. Then

$$\frac{1+e^{-x}}{x} > \frac{1}{x} \geq 0$$

holds for all $x \geq 1$. Since the integral $\displaystyle\int_1^\infty \frac{1}{x}\,dx$ is divergent, we conclude by the comparison theorem that the integral $\displaystyle\int_1^\infty \frac{1+e^{-x}}{x}\,dx$ is also divergent.

## (4) Monotonic function theorem

Attention! *You should understand the Monotonic function theorem*.

We conclude this section with an important theorem, which is needed in the sequel.

Recall that a monotonic function is a function which is either increasing or decreasing. For instance $f(x) = x^2$ is increasing for domain $[0, \infty)$, decreasing for domain $(-\infty, 0]$, and is not monotonic for domain $(-\infty, \infty)$.

Also a bounded function is a function where the function value $y = f(x)$ has a lower bound $m$ and an upper bound $M$. For instance $f(x) = x^2$ is not bounded because no number $M$ can serve as an upper bound.

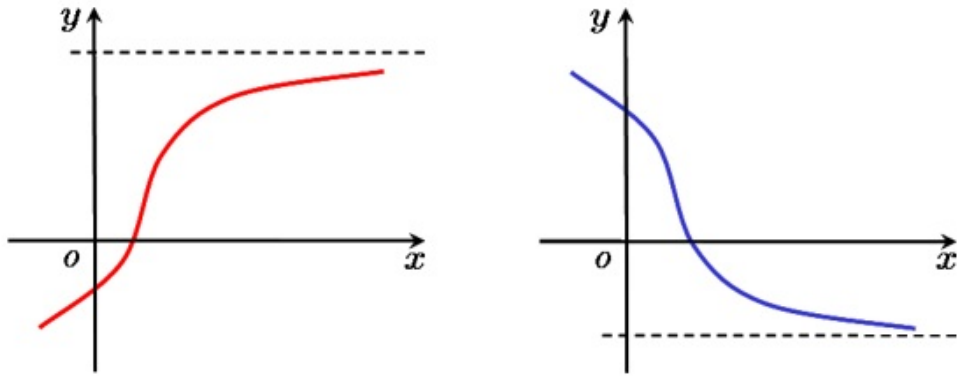
However the function $g(x) = \sin(x^2)$ is bounded because $-1 \leq \sin(x^2) \leq 1$, for all $x$.

## • Monotonic function theorem

If $f(x)$ is a monotonic and bounded function then the limits

$$\lim_{x \to \infty} f(x) \quad \text{and} \quad \lim_{x \to -\infty} f(x)$$

exist and are finite.




(The left figure shows an increasing function with an upper bound, while the right figure shows a decreasing function with a lower bound. In both cases the functions are monotonic and bounded).

# END OF CHAPTER 5